

Humanoid Robots in the Wild: A Digital Ethnography of Online Videos

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ABSTRACT

As consumer humanoid robots become increasingly accessible, human-robot interactions that were once confined to research laboratories are now occurring spontaneously in homes, streets, and public spaces. We present an exploratory digital ethnography analyzing online videos documenting and characterizing naturalistic encounters between humans and the Unitree G1 humanoid robot.

KEYWORDS

humanoid robots, human-robot interaction, digital ethnography, consumer robotics

1 INTRODUCTION

Consumer humanoid robots have rapidly become accessible over the past two years. The Unitree G1, released in 2023 at \$16,000–\$18,000, represents a dramatic shift from research-grade humanoids that typically cost over \$100,000. As owners bring these robots into homes and public spaces, human-robot encounters once confined to laboratories are now documented on social media platforms by robot owners and bystanders.

This shift creates an opportunity to observe naturalistic interaction patterns in the wild, particularly significant because little prior HRI research has involved full-scale, dynamically-balancing humanoids outside controlled laboratory settings. While prior work has examined social robots in homes [4, 10] and public spaces [9], these studies primarily involved wheeled robots like Roombas, stationary systems, or researcher-controlled deployments with pre-determined tasks. Consumer humanoids differ fundamentally: they are bipedal, human-scale, operate with ambiguous autonomy, and enable spontaneous, unscripted human behaviors across diverse environments.

We present a digital ethnography analyzing online videos of human-humanoid interactions. Through systematic coding of interactions across varied settings (streets, homes, retail), we make several observations about how people interact with the robot and how they modify the robot. These observations provide early insights into how humans actually engage with humanoid robots beyond laboratory constraints.

2 METHOD

Digital ethnography is a research method that adapts traditional ethnographic methods to online, technologically-mediated environments, enabling researchers to study communities and practices in digital spaces [2, 6]. It involves searching through unsolicited posts, videos, comments, and other digital artifacts to observe how people naturally represent and reflect on their experiences without researcher prompting. The method has been used in HRI research

previously to observe how users appropriate robots in homes [3], analyze online community discussions about social robots [8], and conduct field-based ethnographies in real-world deployment settings [1].

We adopt the digital ethnography method to better understand ways in which humanoid robots are being used and how people are reacting to encounters with humanoid robots in everyday settings. We focus on videos as the digital medium for our inquiry as they capture rich and candid information about human-robot interactions. Although humanoid robot videos have proliferated in the past two years, we focus on the Unitree G1 robot specifically due to its wider availability to consumers at the relatively low price point of \$16,000. Significantly more user generated videos are available for the Unitree G1, compared to any other functional humanoid.

Our study focuses on video-sharing platforms, including both short and long form videos. To minimize bias in the search results we used Google’s video search, which returns videos across video sharing platforms including YouTube, Reddit, Instagram, Facebook, and X, in the browser’s incognito mode. Our search queries included "unitree humanoid robot" with modifiers such as "in home," "at the mall," "around children," or "walking in downtown." We inspected the top 50 videos from each query, and excluded videos that (1) do not actually involve the Unitree G1, (2) only include company-provided footage, (3) are filmed in controlled laboratory, conference, or tradeshow settings, or (4) do not include natural, unscripted human-robot interactions. After applying these criteria, 60 videos met our inclusion criteria for analysis.

3 FINDINGS

Our analysis reveals several themes and observations in human-humanoid interactions in naturalistic settings.

Proximity Patterns Contradict Safety Guidelines. Of the 60 videos analyzed, proximity distributions revealed systematic close-range engagement: 16 videos (27%) involved touch-distance interactions including handshakes, physical assistance, and direct contact; 26 videos (43%) showed close proximity (<3ft) engagement; 10 videos (17%) captured medium-distance (3-6ft) observation; and 8 videos (13%) showed far-distance (>6ft) encounters.

Notably, 70% of interactions (42/60 videos) occurred at close proximity or touch distance, directly contradicting the manufacturer’s recommendation to maintain 2-meter distance, avoid touching the moving robot, and keep spaces clear. Violations included store employees fitting shoes on robots, operation in cluttered spaces causing collisions and falls, intentional crowd approaches, and children chasing robots at close range. This gap between guidelines and practice suggests that users do not read manuals, perceive small



Figure 1: Examples of robot personalization from our dataset. Robots were customized with clothing and accessories including dashikis, durags, chains, hats, and complete costumes. 28% of videos showed such customization, often accompanied by naming and anthropomorphization.

humanoids as safer than warned, or find guidelines impractical for desired use cases.

Our analysis also demonstrates that prior work in the HRI literature on proxemics [5, 7, 11] might not correctly predict people’s attitudes and behaviors with humanoids in the wild. While proxemics research has established preferred interaction distances of 0.5-1.2m with robots, we observed that 70% of real-world interactions occurred at much closer distances (<3ft/0.9m), often involving physical touch. This systematic violation suggests that humanoid form factor may fundamentally alter proxemic expectations compared to wheeled or non-anthropomorphic robots studied in prior work.

Personalization Despite Functional Limitations. Personalization appeared in 28% of videos (17/60), including accessories (wigs, chains, hats, sneakers) and clothing (dashikis, shirts, durags, costumes). These purely decorative modifications served to anthropomorphize robots or express owner identity, often accompanied by naming (e.g., “Abdul” for a dashiki-wearing robot).

Anthropomorphization Despite Teleoperation. Participants consistently anthropomorphized robots through naming, pronoun shifts (it to he), verbal commands, and social rituals despite knowing they were teleoperated. For example, one user named their robot “Abdul” and said “he means well” when stuck, suggesting humanoid form alone triggers social treatment. Bystanders exhibited varied reactions—fear, amusement, curiosity—with robots used predominantly for entertainment rather than functional tasks.



Figure 2: Examples of ways people interact with the Unitree G1 robot in close proximity and physically touch it (holding or shaking hand, pushing torso, putting on a shoe).

4 DISCUSSION

Are humanoid robots useful and acceptable? Our dataset shows limited functional utility—most interactions centered on entertainment rather than practical tasks. However, 70% close-proximity engagement suggests comfort with physical interaction, indicating cautious acceptability despite some initial bystander fear. **Are humanoid robots acceptable?** The data suggests cautious optimism. While some bystanders showed initial fear, most exhibited curiosity or indifference. The 70% close-proximity interaction rate indicates comfort with physical engagement. However, our dataset captures predominantly outdoor encounters; long-term domestic acceptability remains uncertain.

What do these findings suggest for design? First, the 28% personalization rate indicates manufacturers should provide official customization options that maintain sensor functionality. Second, consistent violation of 2-meter guidelines suggests safety protocols need revision—designers should assume human contact as default and implement collision avoidance, force-limiting actuators, and compliant materials. Third, persistent anthropomorphization despite known teleoperation suggests leveraging social affordances over functional utility.

What are the limitations? Our dataset focuses heavily on the Unitree G1, analyzed only public videos, spans only 2024-2025, and features primarily teleoperated rather than autonomous behavior, potentially limiting generalizability.

What future work does this enable? Comparative studies across models and contexts, systematic investigation of the safety guideline gap, controlled experiments validating observations, and continued digital ethnography tracking interaction evolution.

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